# Research Proposal: Evaluating the Impact of Augmenting the VAEP Model with 360-Degree Freeze-Frame Spatial Context

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## Table of Contents

1. Introduction ..... 2
2. Research Objectives ..... 3
3. Background ..... 3
    3.1. Valuing Actions by Estimating Probabilities (VAEP) ..... 3
4. Methodology ..... 7
    4.1. Datasets ..... 7
        4.1.1. Limitations of Datasets ..... 7
    4.2. Spatial Context ..... 8
        4.2.1. Feature Engineering ..... 8
        4.2.2. Embeddings ..... 8
    4.3. VAEP Model Training ..... 8
        4.3.1. Logistic Regression ..... 8
        4.3.2. Gradient Boosting ..... 8
        4.3.3. Loss Functions ..... 9
    4.4. VAEP Model Evaluation Metrics ..... 9
        4.4.1. Brier Score ..... 9
        4.4.2. AUC-ROC ..... 9
        4.4.3. Cross-Entropy Loss ..... 9
5. Project Planning ..... 9
    5.1. Estimated Timeline ..... 9
    5.2. Prerequisite Resources ..... 9
6. Conclusion ..... 9
References ..... 10

## 1. Introduction

The adoption of analytics in association football has grown significantly in the last couple of decades (Goes et al., 2021; Olthof & Davis, 2025; Schaerlaeckens, 2025; Wakelam et al., 2022), enabling football clubs to leverage insights to inform tactical decisions, coaching and recruitment. Much of this innovation has involved the development of metrics used to determine the value provided by a particular action made by a player in terms of goal-scoring probabilities. One such metric is Valuing Actions by Estimating Probabilities (VAEP) (Decroos et al., 2019).

VAEP considers the probability of scoring or conceding within the next $k$ events to assign a value between $-1$ and 1 to each action. VAEP, like other such metrics, typically uses action-centric features such as the action's start and end location and type (carry, pass, shot) in estimating scoring/conceding probabilties, but this approach does not explicitly model the wider spatial context (Sá-Freire, 2025). The spatial context - the position of teammates and opponents around the actor - of an action determines the options available to a player, showing viable passing options or space to carry into, as well as the proximity of opposition defenders applying pressure or blocking options.

The StatsBomb Open Data repository (StatsBomb, 2026) provides events (e.g. carries, passes, shots) data for matches, along with *three-sixty* freeze-frames which indicate the positions of players within proximity of the acting player. Such data can be used to incorporate spatial context as a predictive feature. To illustrate this, **Figure 1** compares a pass event from a match with and without the surrounding spatial context. Without the spatial context, this pass appears to offer limited progression towards the opponent's goal. Traditional VAEP would consider how passes of this trajectory tend to result in scoring/conceding, but the addition of spatial context differentiates this particular pass from others of similar trajectory. **Figure 1** shows how the left hand side of the actor is blocked by multiple opponents (red) and this pass moves the ball into space on the right flank to open an attacking opportunity (in this case, a long-ranged shot)[1]. Spatial context could allow finer-grained differentiation, rather than aggregating all passes of this trajectory.

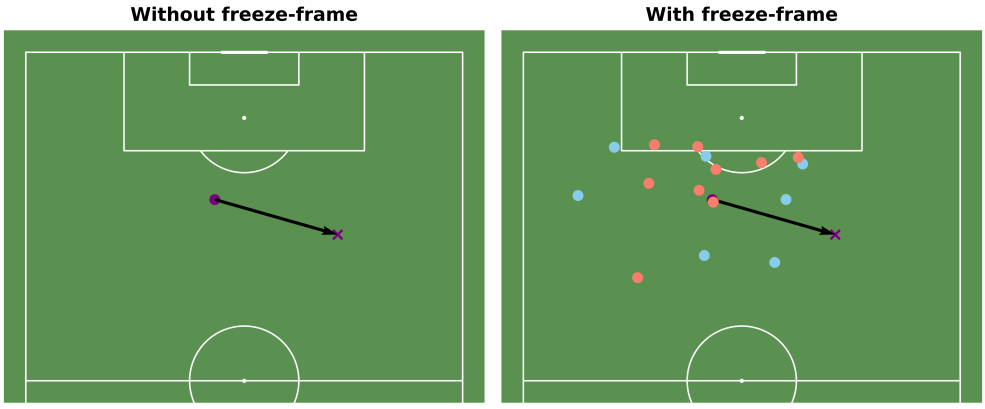



Figure 1: Pass Event With and Without Freeze-Frame

[1]The pass event id is '23b6b26a-293c-4d5e-9889-5a5ce0b7ad8c' and the subsequent long-ranged shot event id is 'd9d27d20-35bc-42bc-ac53-059ad519f781'

## 2. Research Objectives

The general aim of this proposed project is to determine to what extent the addition of spatial context features enhances the standard VAEP model. This entails the following objectives:

- Develop representations of spatial context from StatsBomb 360 freeze-frame data using feature engineering and deep learning techniques.
- Train and evaluate VAEP models, both with and without the inclusion of spatial context representations, using supervised classification algorithms.
- Compare and interpret these VAEP models using appropriate peformance metrics and an exploration of feature importance.

## 3. Background

In soccer analytics, *actions* refer to a *type of event* within a match which is caused by a player (e.g. a pass), with *events* being the broader set of significant occurrences in a match (e.g. passes, but also substitutions, fouls etc.) (Decroos, Bransen, Van Haaren, & Davis, 2019). Typical performance metrics focus on action outcomes, such as goals, assists and pass completion rates. Whilst such metrics can be used to determine the value of some actions and are useful for summarising player ability, soccer analytics has developed metrics to help generalise this notion of action value. Moreover, models used to assign value to actions have been enhanced by incorporating the locations of teammates and opponents, leading to improved performance and interpretability.

As such, this section will introduce the VAEP metric and outline how spatial context has been successfully incorporated into other action valuation metrics, motivating this proposed project's research objectives.

### 3.1. Valuing Actions by Estimating Probabilities (VAEP)

Decroos, Bransen, Van Haaren, & Davis (2019) introduce Valuing Actions by Estimating Probabilities (VAEP). VAEP values actions in terms of the probability of scoring and the probability of conceding within the next $k$ actions. Equation 1 defines the VAEP of an action $a_i$ for team $x$.

$$V(a_i, x) = \Delta P_{scores}(a_i, x) - \Delta P_{concedes}(a_i, x) \tag{1}$$

If we consider a match as a sequence of $m$ on-ball actions $[a_1, ..., a_m]$, we can view some game state $S_i$ which corresponds to some subset of actions $[a_1, ..., a_i]$ and determine how the last action in the game state $a_i$ changed the probability of team $x$ scoring and conceding within the next $k$ actions. The change in the probability of scoring (and conceding) from the previous game state $S_{i-1} = [a_1, ..., a_{i-1}]$ to the current game state $S_i$ can be determined by simply finding their difference:

$$\begin{aligned} \Delta P_{\text{scores}}(a_i, x) &= P_{\text{scores}}(S_i, x) - P_{\text{scores}}(S_{i-1}, x) \\ \Delta P_{\text{concede}}(a_i, x) &= P_{\text{concede}}(S_i, x) - P_{\text{concede}}(S_{i-1}, x) \end{aligned} \tag{2}$$

These two probabilities are estimated using two supervised classification algorithms, one which estimates scoring, and the other conceding, within the next $k$ actions. The feature sets used across these two models are identical, all that varies is the labels (scoring or conceding). Decroos, Bransen, Van Haaren, & Davis (2019) used a feature set which corresponds to the Soccer Player Action Description Language (SPADL) format involving 9 features, most notably the action's start and end locations, and the type of action. The authors also add "complex features", such as the time

elapsed between consecutive actions, as well as "game context features", such as the scoreline at the time of the action and the amount of time remaining.

Decroos, Bransen, Van Haaren, & Davis (2019) initially compared logistic regression, a random forest, XGBoost and CatBoost, finding CatBoost to perform best across both the scoring and conceding models in terms of the Brier Score and the Area Under the Receiver Operating Characteristic Curve (AUC-ROC). Decroos, Bransen, Van Haaren, & Davis (2019) also extend the VAEP metric by using it to provide player ratings, where the VAEP of each action in a player's set of actions $A_p^T$ across some time frame $T$ are considered cumulatively, and normalised per-90 minutes played during $T$.

$$\text{rating } (p) = \frac{90}{m} \sum_{a_i \in A_p^T} V(a_i) \tag{3}$$

The authors compare these VAEP player ratings

The addition of spatial context has been successfully applied to related action value metrics…

Expected goals (xG) is an approach commonly used for assigning value to individual shots (Green, 2012; Macdonald, 2012). General to xG approaches is the representation of the value of some shot as the probability that it results in a goal. These goal probabilities are estimated via various classification algorithms such as logistic regression, gradient boosting and deep learning methods (Mead et al., 2023). Features used in xG models typically include *action-intrinsic* variables such as distance/angle to the centre of the opponent's goal and the body part used, along with *match-context* variables such as the current match scenario (e.g. set piece, open play) (Anzer & Bauer, 2021; Mead, O'Hare, & McMenemy, 2023; Robberechts & Davis, 2020).

Cefis & Carpita (2025) incorporate spatial context into the xG model by considering the number of players ahead of the shooter ($x$ co-ordinate greater than the shooter's) within a fixed radius ($y$ co-ordinate $\pm 10\%$ of the shooter's). From this count, they derive two binary variables `D1.OpponentsPlayer` and `D2.OpponentsPlayer`, where the former equals 1 when the number of players in this area is greater than 0, and the latter equals 1 when the number of players in the area is greater than 1. They also consider the $x$ and $y$ co-ordinates of the opponent's goalkeeper (`GK x coordinate`, `GK y coordinate`). Table 3 in Cefis & Carpita (2025) show the coefficients and odds ratios of each feature in their logistic regression xG model across their 4 different techniques for addressing class imbalance. `D1.OpponentsPlayer`, `D2.OpponentsPlayer`, and `GK x coordinate` all have negative coefficients, indicating a reduction in the log odds of the shot resulting in a goal. This is particularly pronounced for `D2.OpponentsPlayer`, with a coefficient range of $-6.13$ to $-5.69$, indicating that the probability of scoring (i.e. xG) is reduced by around 99.73% (because $1 - e^{(-5.69+-6.13)/2} \approx 0.9973$) when there is more than 1 opposition defender within proximity of the shooter. This finding matches intuitions about the influence of the opposition's positioning on goal probability and highlights how the addition of spatial context dureing modelling can enhance the xG metric.

A further action value metric is expected threat (xT). xT measures the value of actions based on where on the pitch the action moves the ball from and to. Singh (2018) derived the following equation for calculating the xT of a given location on the pitch:

$$\text{xT}_{x,y} = \left(s_{x,y} \times g_{x,y}\right)\left(m_{x,y} \times \sum_{z=1}^{Z}\sum_{w=1}^{W} T_{(x,y)\to(z,w)}\text{xT}_{z,w}\right) \tag{4}$$

If we discretise the football pitch into some $Z \times W$ (Singh (2018) uses $Z = 16$ and $W = 12$) set of equally-sized square zones $(x, y)$, we can consider three possible attacking options when the

ball is in some zone $(x, y)$: shoot, carry or pass. $\left(s_{x,y} \times g_{x,y}\right)$ quantifies the value of shooting from zone $(x, y)$ by weighting the probability of scoring $g_{x,y}$ by the probability of shooting $s_{x,y}$. The other two options, carrying or passing, involve moving the ball from $(x, y)$ to $(z, w)$. The transition matrix $T_{(x,y)}$ provides the probablity of moving the ball from zone $(x, y)$ to any zone $(z, w)$ after 1 time step. If we consider all $Z \times W$ possible options and multiply their transition probabilities by the value of its respective $(z, w)$ zone, we can take their sum and weight it by the probability of actually moving the ball to another zone when in $(x, y)$. This provides a quantification of passing or carrying the ball, which when multiplied by the shot-probability weighted xG of zone $(x, y)$ provides the xT of the ball being at zone $(x, y)$. However, given that Equation 4 assumes that the value of each zone $\left(\mathrm{xT}_{z,w}\right)$ is already known a-priori, we must estimate its value. The xT model thus begins by setting every zone $(x, y)$ to have an xT of 0. This simplies Equation 4 to essentialy be the xG of zone $(x, y)$, weighted by how likely players are to actually shoot from that zone:

$$\mathrm{xT}_{x,y} = \left(s_{x,y} \times g_{x,y}\right) \tag{5}$$

We can then take the xT values from this first iteration and recalculate them iteratively until convergence. Singh (2018) suggests 4–5 iterations are sufficient but that this may vary across datasets. To then assign xT to an action, we simply subtract the xT of the end location of the action (i.e. the xT of zone $(z, w)$) by the xT of the start location (the xT of zone $(x, y)$). This approach is useful for allowing us to determine the value of any on-ball attacking action, not just shots. We can also quantify genuinely valuable actions at any point during a chain of actions (e.g. an incisive pass during a sequence which ends in a goal), rather than suffering from a bias towards teminating actions (goals, assists).

Everett et al. (2022) introduce spatial context to the xT transition matrix $T_{(x,y)}$ via a pitch control (Spearman et al., 2017) and interception model (Burriel & Buldú, 2021) which was used to train a convolutional neural network (CNN) to produce the transition probabilities for their contextual transition matrix $T^{\star}_{(x,y)}$. The pitch control model considers the probability of a player controlling the ball as a function of their distance from opposition players, whilst the interception model introduces an additional transition probability, that being the probability of the ball being intercepted when moving from zone $(x, y)$ to its intended destination $(z, w)$. Everett, Beal, Matthews, Norman, & Ramchurn (2022) found the introduction of spatial context yielded a modest improvement in terms of model fit, reducing the log likelihood of the xT model from $-5.137$ to $-4.151$. Greater gains were found in its useful applications to tactical analysis, such as enabling the authors to apply a Basin-Hopping optimiser (Wales & Doye, 1997) to find more optimal defensive shapes than which actually occurred during actions, using their derived Threat Above Expected (TAx) metric as the algorithm's loss function.

# 4. Methodology

## 4.1. Datasets

The proposed project will use data from StatsBomb's Open Data repository. This repository has "made certain leagues of StatsBomb Data freely available for public use for research projects and genuine interest in football analytics" (StatsBomb, 2026). The StatsBomb Open Data repository provides the following datasets, and data specification documents exist which outline variables, data types and variable descriptions for each dataset:

- **competitions.json:** A single JSON file for competition summary information.
- **events:** A directory containing a single JSON file for each match, detailing each event which takes place within that match (e.g. each pass, shot, carry etc.).
- **lineups:** A directory containing a single JSON file for each match, providing information about each player who participated in that match (e.g. their position, shirt number, nationality).
- **matches:** A directory containing a single JSON file for a season in a given competition, providing summary information about each match played in that season (e.g. match date, home/away teams and scores).
- **three-sixty:** A directory containing a single JSON file for each match, providing data for each event in a given match detailing the position of each player in the broadcaster's frame at the moment of that event.

Decroos, Bransen, Van Haaren, & Davis (2019), along with VAEP, introduce the Soccer Player Action Description Language (SPADL) as a data engineering format to enable "human-interpretable, simple and complete" records of actions within football matches. Such a format will be used in this proposed project for the baseline feature set. This format will then be augmented by this proposed project's engineered spatial features.

The proposed project will use *events* and *three-sixty* data corresponding to 292 matches from UEFA Men's Euro 2020 & 2024, UEFA Women's Euro 2022 & 2025, FIFA Men's World Cup 2022, FIFA Women's World Cup 2023. Unlike most competitions in this repository, *three-sixty* data is available for every match in this sample, enabling player positional context to be incorporated into feature engineering for 1,052,116 match events.

### 4.1.1. *Limitations of Datasets*

Football analysts often makes use of tracking data (Goes, Meerhoff, Bueno, Rodrigues, Moura, Brink, Elferink-Gemser, Knobbe, Cunha, Torres, & Lemmink, 2021). Rather than considering discrete time events, tracking data provides a continuous record of player and ball position for the duration of a match. This enables analysts to model actions in matches as continuous processes, accounting for the position and movements of the entire team, unlike with events data. Such tracking data is often proprietory and is thus not available for public use (Bassek et al., 2025). This limits the scope of the present project to viewing events as discrete in time, thus losing some surrounding context which may be informative (e.g. player movement direction and velocity).

Events data is nonetheless relatively comprehensive, covering a much larger sample of matches than the limited publicly-available tracking data (Olthof & Davis, 2025). Furthermore, the use of StatsBomb's *three-sixty* data does address this limitation somewhat by providing snapshots of proximal player positions at the point of each event, allowing some inference to be made regarding team positioning during match events. However, these snapshots do not necessarily capture every player's location during an event, just those within the view of the broadcast camera (StatsBomb, 2026).

## 4.2. Spatial Context

### 4.2.1. *Feature Engineering*

### 4.2.2. *Embeddings*

```
{
    "teammate": boolean,
    "actor": boolean,
    "keeper": boolean,
    "location": [float, float]
}
```

Figure 2: StatsBomb *three-sixty* JSON schema

As seen in Figure 2, StatsBomb *three-sixty* data does not identify players within each freeze frame. Thus, to create embeddings for player positions, we must use a neural network which has an input layer which is *permutation invariant*. That is, the output of the network (i.e. the embeddings) should not depend on the order in which features appear in the input layer.

Zaheer et al. (2018) introduce *Deep Sets* as a framework for using *sets* as inputs to a network used to create a $d$-dimensional embedding of that set.

$$f(X) = \rho\left(\sum_{x \in X} \phi(x)\right) \tag{6}$$

## 4.3. VAEP Model Training

### 4.3.1. *Logistic Regression*

$$P(Y = 1 \mid X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n)}} \tag{7}$$

L2 Regularisation Limited-memory BFGS

### 4.3.2. *Gradient Boosting*

$$\hat{y_i} = \sum_{t=1}^{T} f_t(x_i), \quad f_t \in \mathcal{F} \tag{8}$$

#### 4.3.2.1. *Random Forest*

100 trees 40 parallel threads

#### 4.3.2.2. *XGBoost*

100 trees maximum depth 3 learning rate 0.1 40 parallel threads

#### 4.3.2.3. *Catboost*

40 parallel threads

#### 4.3.3. *Loss Functions*

##### 4.3.3.1. *Cross-Entropy Loss*

$$H(p, q) = -\sum_{x \in X} p(x) \log q(x) \tag{9}$$

##### 4.3.3.2. *Gini impurity*

$$G_m = 1 - \sum_{k=1}^{K} \left( \frac{n_{m_k}}{n_m} \right)^2 \tag{10}$$

### 4.4. VAEP Model Evaluation Metrics

#### 4.4.1. *Brier Score*

$$\mathrm{BS} = \frac{1}{N} \sum_{j=1}^{r} \sum_{i=1}^{n} \left( f_{ij} - E_{ij} \right)^2 \tag{11}$$

For binary classification this simplifies to:

$$\mathrm{BS}_{\mathrm{binary}} = \frac{1}{N} \sum_{i=1}^{N} \left( p_i - y_i \right)^2 \tag{12}$$

#### 4.4.2. *AUC-ROC*

#### 4.4.3. *Cross-Entropy Loss*

## 5. Project Planning

### 5.1. Estimated Timeline

- A Gantt Chart

### 5.2. Prerequisite Resources

- Python Version: 3.12.10
  - Statsbombpy, Socceraction, mplsoccer, Statsmodels, Scikit-Learn, PyTorch
- Positron, Quarto, Typst
- My hardware (CPU, GPU, RAM)

## 6. Conclusion

## References


Anzer, G., & Bauer, P. (2021). A Goal Scoring Probability Model for Shots Based on Synchronized Positional and Event Data in Football (Soccer). *Frontiers in Sports and Active Living*, *3*, 624475–624476. https://doi.org/10.3389/fspor.2021.624475

Bassek, M., Rein, R., Weber, H., & Memmert, D. (2025). An Integrated Dataset of Spatiotemporal and Event Data in Elite Soccer. *Scientific Data*, *12*(1), 195–196. https://doi.org/10.1038/s41597-025-04505-y

Burriel, B., & Buldú, J. M. (2021). The Quest for the Right Pass: Quantifying Player's Decision Making. *Proceedings of the StatsBomb Innovation in Football Conference*, 1–22. https://statsbomb.wpengine.com/wp-content/uploads/2021/11/Javier-M-Buldu.pdf

Cefis, M., & Carpita, M. (2025). A New xG Model for Football Analytics. *Journal of the Operational Research Society*, *76*(1), 1–13. https://doi.org/10.1080/01605682.2024.2323669

Decroos, T., Bransen, L., Van Haaren, J., & Davis, J. (2019). Actions Speak Louder than Goals: Valuing Player Actions in Soccer. *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 1851–1861. https://doi.org/10.1145/3292500.3330758

Everett, G., Beal, R., Matthews, T., Norman, T., & Ramchurn, G. (2022). Contextual Expected Threat Using Spatial Event Data. *Statsbomb Conference 2022*. https://geverett.co.uk/pdfs/Gregory-Everett-et-al-Contextual-xT-using-Spatial-Event-Data.pdf

Goes, F., Meerhoff, L., Bueno, M., Rodrigues, D., Moura, F., Brink, M., Elferink-Gemser, M., Knobbe, A., Cunha, S., Torres, R., & Lemmink, K. (2021). Unlocking the Potential of Big Data to Support Tactical Performance Analysis in Professional Soccer: A Systematic Review. *European Journal of Sport Science*, *21*(4), 481–496. https://doi.org/10.1080/17461391.2020.1747552

Green, S. (2012). *Assessing The Performance of Premier League Goalscorers*. https://www.statsperform.com/resource/assessing-the-performance-of-premier-league-goalscorers/

Macdonald, B. (2012). An Expected Goals Model for Evaluating NHL Teams and Players. *Proceedings of the 2012 MIT Sloan Sports Analytics Conference*, }, % month={March.

Mead, J., O'Hare, A., & McMenemy, P. (2023). Expected Goals in Football: Improving Model Performance and Demonstrating Value. *PLOS ONE*, *18*(4), e282295. https://doi.org/10.1371/journal.pone.0282295

Olthof, S., & Davis, J. (2025). Perspectives on Data Analytics for Gaining a Competitive Advantage in Football: Computational Approaches to Tactics. *Science and Medicine in Football*, *0*(0), 1–13. https://doi.org/10.1080/24733938.2025.2533784

Robberechts, P., & Davis, J. (2020). *How Data Availability Affects the Ability to Learn Good xG Models* (pp. 17–27). https://doi.org/10.1007/978-3-030-64912-8_2

Schaerlaeckens, L. (2025). The Belgian Lab Shaping Modern Soccer's Data Revolution. *The Guardian*. https://www.theguardian.com/football/2025/jun/03/soccer-analytics-jesse-davis-leuven

Singh, K. (2018). *Introducing Expected Threat (xT)*. https://karun.in/blog/expected-threat.html

Spearman, W., Basye, A., Dick, G., Hotovy, R., & Pop, P. (2017, March 3). *Physics-Based Modeling of Pass Probabilities in Soccer*. https://www.researchgate.net/publication/315166647_Physics-Based_Modeling_of_Pass_Probabilities_in_Soccer

StatsBomb. (2026). *StatsBomb Open Data* [Computer software]. StatsBomb. https://github.com/statsbomb/open-data

Sá-Freire, L. M. de. (2025). Event Data Based Action Valuation: Possibilities and Limitations. In D. Memmert (Ed.), *Artificial Intelligence and Machine Learning in Sports Science* (pp. 285–292). Springer. https://doi.org/10.1007/978-3-662-70155-3_20

Wakelam, E., Steuber, V., & Wakelam, J. (2022). The Collection, Analysis and Exploitation of Footballer Attributes: A Systematic Review. *Journal of Sports Analytics*, *8*(1), 31–67. https://doi.org/10.3233/JSA-200554

Wales, D. J., & Doye, J. P. K. (1997). Global Optimization by Basin-Hopping and the Lowest Energy Structures of Lennard-Jones Clusters Containing up to 110 Atoms. *The Journal of Physical Chemistry a*, *101*(28), 5111–5116. https://doi.org/10.1021/jp970984n

Zaheer, M., Kottur, S., Ravanbakhsh, S., Poczos, B., Salakhutdinov, R., & Smola, A. (2018, April 14). *Deep Sets*. https://doi.org/10.48550/arXiv.1703.06114